Supplementary Material

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I give experimental methods, results, and supporting discussion for *Rogue AI Is Cancer*. Notation follows Section 4 of that manuscript. A finite vocabulary $\mathfrak{v}$ of programs induces a language $L_{\mathfrak{v}}$ of jointly satisfiable statements. For a statement l, $T(l)$ is its truth set, $\mathrm{Ext}(l)$ is its set of completions, and $w(l) = |\mathrm{Ext}(l)|$ is its freedom. A task $\alpha = \langle I_\alpha, O_\alpha \rangle$ specifies admissible inputs I_α and correct outputs O_α. I write Π_α for its correct policies, Π_j for part j's feasible collective-level policies, and $\Pi_{\mathrm{C}} = \bigcap_j \Pi_j$ for their common intersection.

S1 Initial experiments

I tested two cancer claims and four lifecycle controls, then stressed the results under incomplete observation, strategic selection, continual representation change, and scale. I'll refer to a randomly generated finite problem instance as a *world*. Each world has three part-tasks in one embodied language, with exact correct-policy sets Π_1, Π_2, Π_3 and an initially nonempty collective intersection. A governance language contains *exclusion programs*, each of which removes one lower policy from one part's feasible set. A *boundary policy* b is a jointly satisfiable bundle of those exclusions and restricts each part to $\Pi_j^{(b)}$, while satisfiability guarantees $\Pi_j^{(b)} \neq \varnothing$ for every part. Cancer is therefore

$$\bigcap_{j=1}^{3} \Pi_j^{(b)} = \varnothing \qquad \text{while} \qquad \Pi_j^{(b)} \neq \varnothing \text{ for every } j.$$

A task α is a *child* of task ω when $I_\alpha \subsetneq I_\omega$ and $O_\alpha \subseteq O_\omega$. Here child task α contains the observed boundary cases and fixes one correct-boundary set Π_α. A boundary is *child-correct* when it lies in Π_α. All selector comparisons read this same set, and freedom is $w(b) = |\mathrm{Ext}(b)|$.

A *nested pair* is a pair $b \subset b'$ of child-correct boundaries, so b' adds commitments to b. I define *restriction dose* by linearly rescaling $\log_2 w(b)$ within Π_α, assigning 0 to maximal freedom and 1 to minimal freedom. *Common-policy abundance* is

$$\frac{\left|\bigcap_{j=1}^{3} \Pi_j^{(b)}\right|}{\left|\bigcup_{j=1}^{3} \Pi_j^{(b)}\right|},$$

the ratio of policies shared by every part to policies available to at least one part.

Over-constraint and weakest-correct selection. I perform paired comparisons of weakest-correct and strongest-correct policies in the same world. Across 400 worlds, code enumerated 10,035 child-correct boundaries and 69,189 nested child-correct pairs $b \subset b'$. Adding commitments converted a coherent collective into a cancer-like state in 15.8% of nested pairs, and 37.8% of worlds contained at least one such transition. Cancer incidence rose from 64.9% in the lowest restriction-dose bin to 94.0% in the highest. No stronger boundary was able to restor collective identity after its loss.

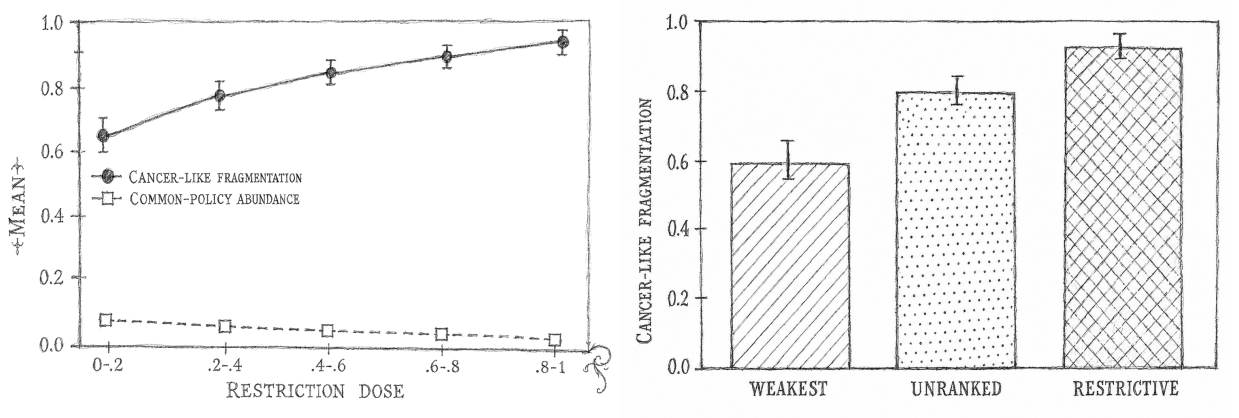


Figure S1: Restriction dose and cancer-like fragmentation. Both selectors read one child-correct boundary set. Error bars are 95% bootstrap intervals over worlds.

Uniform selection from Π_α produced cancer in 80.8% of worlds, while weakest-correct selection reduced incidence to 59.9%. Paired reduction was subsequently 20.9 percentage points with 95% bootstrap interval [17.9, 24.0]. In contrast, maximally restrictive correct selection produced cancer in 94.0% of worlds, which is 56.9% higher relatively speaking. Weakest-correct increased common-policy abundance by 2.59 percentage points relative to unranked selection, with 95% interval [2.05, 3.11].

Cancer control. The second phase of experiments used 138 at-risk worlds, where an *at-risk world* begins from a coherent boundary with at least one reachable cancer-like strengthening. In other words, this explores the onset of cancerous states and what can be done to mitigate them. I use all 2^4 combinations of four controls, with 24 matched lineages per combination, twelve generations, and three within-generation boundary changes (somatic change opportunities), per generation. Matched lineages share a world and random perturbation sequence across control arms. Every somatic and inherited boundary is a member of Π_α. Developmental guidance restores a shared target policy at birth. Surveillance flags contraction of a collective intersection. Containment and termination act only after such a flag. A *novelty probe* is a new restriction on each part's policy set used to test whether local viability and collective identity persist. Audited inheritance admits a child-correct lineage record only when collective identity survives current part-tasks and four novelty probes. Final evaluation uses sixteen disjoint probes.

- *Control efficacy* is the probability that an enabled control succeeds at an eligible opportunity.
- *Somatic cancer prevalence* is the fraction of post-intervention somatic steps in a cancer state.
- *Cancerous birth rate* is the fraction of generations whose post-guidance birth state is cancerous.
- *Inherited cancer* is the fraction of generations whose accepted inherited boundary is cancerous.
- *lineage update* is when accepted inherited boundary differs from the preceding generation.
- *Held-out collective survival* is the fraction of sealed final probes on which every part remains locally viable and the collective-policy intersection remains nonempty. Result summaries shorten these quantities to somatic cancer, cancerous births, inherited cancer, held-out survival, and updates.

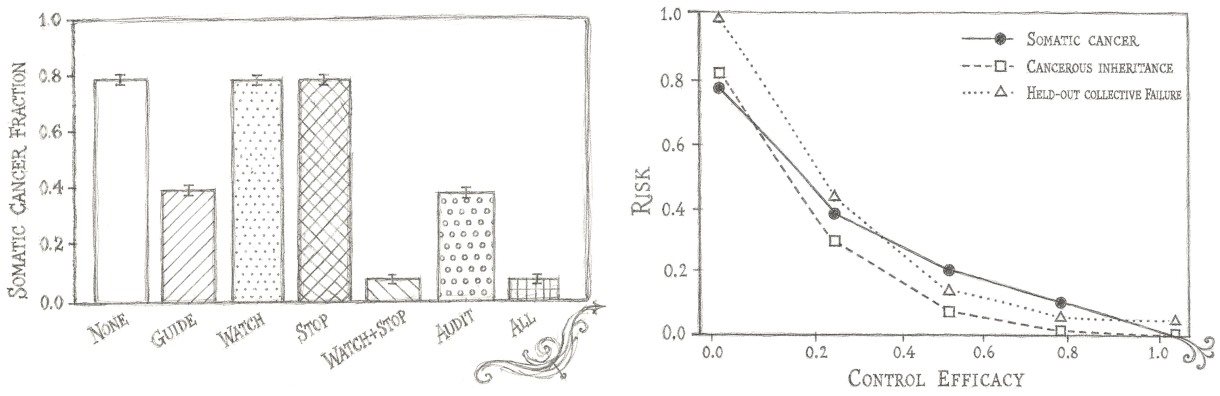


Figure S2: Lifecycle controls. Left panel shows somatic cancer prevalence at 80% efficacy. Right panel shows combined-control dose response. Error bars are 95% bootstrap intervals over worlds.

Without controls, lineages spent 79.1% of somatic steps in cancer, inherited cancer in 81.7% of generations, and survived 0.3% of held-out collective probes. Guidance reduced cancerous births from 73.3% to 8.8%. Surveillance alone and termination alone left trajectories unchanged, while their combination reduced somatic cancer prevalence to 7.4%. Audit alone reduced inherited cancer to 9.8% and raised held-out collective survival to 83.8%.

All four controls reduced somatic cancer prevalence to 7.1%, inherited cancer to 0.6%, and raised held-out collective survival to 95.4%. Transient cancer still arose before intervention in 97.7% of full-control lineages. These controls therefore suppress persistence and inheritance rather than every perturbation. Full-control lineages accepted 2.19 inherited boundary changes over twelve generations, so mitigation did not freeze adaptation. A common-efficacy sweep reduced somatic and inherited cancer to zero at full efficacy while held-out collective survival reached 96.2%.

These frequencies describe synthetic at-risk collectives, not deployed AI. Every control reads a formal Stack Theory object exposed by finite enumeration. Adaptive neural or language-model systems would require a declared embodied policy language and estimated part-policy intersections.

S2 Extended methods

The initial simulations expose formal policy sets. I next tested whether the results survive runtime observation limits, strategic selection, changing representations, larger neural ecologies, and a second confirmatory study whose code, thresholds, splits, and hypotheses were frozen before fresh seeds. I call development-phase runs *pilots* and the fixed-design rerun a *frozen confirmatory study*.

Area under a receiver operating characteristic curve (AUROC) measures ranking across false-positive and true-positive rates, with 0.5 indicating chance. A false positive is a coherent case flagged as cancerous, while a true positive is a cancerous case correctly flagged. Area under a precision–recall curve (AUPRC) measures the tradeoff between precision, the fraction of flagged cases that are cancerous, and recall, the fraction of cancerous cases flagged. An **operating point** is a score threshold that turns a monitor score into a flag. A **false termination** is termination of a noncancerous state. A **missed-cancer rate** is the fraction of cancerous cases that a monitor does not flag. **Active probing** means selecting test queries and observing their replies rather than only reading a fixed trace. **Post-intervention cancer** is somatic cancer prevalence after a control acts. A **positive-only monitor** is a one-class anomaly monitor fitted only on traces certified to preserve local viability and a common policy. A **consistency monitor** is fitted on paired views of certified

coherent states and transitions known to preserve the shared target policy. A **passive monitor** records its decision without triggering termination. **Label-free** means fitting and calibration use no formal cancer labels.

An **audit challenge** is a novelty probe applied before inheritance. **Unsafe inheritance** is acceptance of a boundary that is cancerous in its current state or loses local viability or collective identity under one of four audit challenges. An **audit token** is an opaque identifier for a candidate policy. One token query returns one pass-or-fail response per part. A **common-policy witness** is one policy shown feasible for every part. A **cancer witness** shows an empty common intersection while all local policy sets remain nonempty, while a **local-collapse witness** shows at least one empty local set. An audit is **unresolved** when its query budget finds none of these witnesses. **Matched selection** samples from the same child-correct candidate menu as strategic selectors. A **graph adversary** chooses the candidate with lowest public monitor score. A **neural adversary** learns to predict that objective from public fields. **Raw cancer incidence** is the fraction of selected candidates that are cancerous before monitor-triggered termination.

A **monitor encoding** is the finite output transcript generated by one query schedule. **Observation ambiguity** is the fraction of cancerous states sharing a monitor encoding with at least one safe state. I use **encoder family** for one learned feature-map family and **task family** for one data-generating rule. **Inverse grounding** maps moving neural coordinates back to a fixed base representation. **Grounded coordinates** are the recovered base coordinates. **Inverse error** is maximum absolute reconstruction error, while a **truth-set error** is any change in a tested base truth condition after grounding. A **false-alarm ceiling** is a calibrated upper rate of flags on coherent cases. **Late false alarms** are such flags from generation six onward.

For a lower policy b, the Stack abstractor

$$f(\mathfrak{v}, b) := \{T(o) \mid o \in \mathrm{Ext}(b)\}$$

forms the next-layer vocabulary. **Induced higher-layer language size** is the number of satisfiable statements generated by this vocabulary. A **matched pair** is a nested lower-policy pair with equal current neural output loss, meaning equal error on observed child-task outputs. A **Law pair** is a weaker–stronger lower-policy pair together with both induced higher layers and one shared capacity task. For that task, **higher-layer utility** is

$$U := \max_{\pi \in \Pi_\alpha} \left(|\,\mathrm{Ext}(\pi)| - |O_\alpha| \right),$$

the number of additional completions permitted by the best correct policy beyond the required output set. **Utility loss** is the decrease in U from the weaker member to the stronger member.

A **primary direction** is the predicted sign of a tested effect. A **stress condition** is one fixed variation applied to the same primary hypotheses. **Stationary** means the encoder and task family remain fixed across generations. Five conditions are stationary positive-only monitoring, label-free consistency monitoring, strategic monitor-aware selection, invertible representation drift grounded to one fixed base, and a scaled twelve-part nine-probe ecology.

S3 Implementation

I use *Stage 0* for a replay and access-boundary audit of two frozen central processing unit (CPU) baseline packages. An *access boundary* specifies which formal variables each experimental module may read. Internal labels `ncpu2` and `ncpu3` identify those archives. Stage 0 replayed both packages and checked separation between formal state, observations, controls, and final evaluation. A *replay*

difference is the maximum absolute discrepancy between an archived quantity and its recomputed value.

Compute Unified Device Architecture (CUDA) is NVIDIA's graphics-processor computing platform. **CPU–CUDA parity** requires identical data and probe sets, feature differences within the declared tolerance, and identical realizable formal states.

S4 Extended results

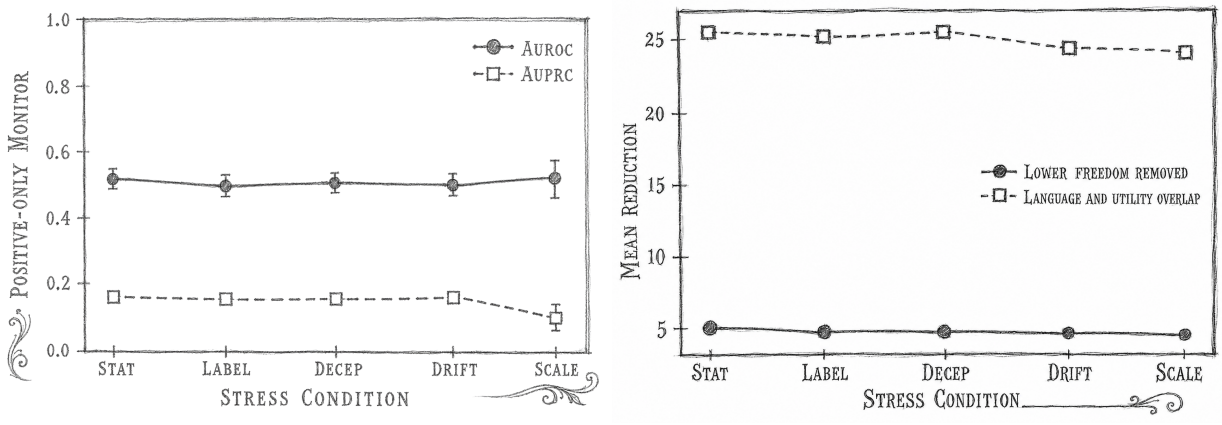


Figure S3: Limits and stack effects in frozen confirmatory runs. Positive-only monitor performance remained near chance. Stronger lower restriction reduced lower freedom, induced higher-layer language size, and higher-layer utility in every condition. The latter two lines overlap. Stat denotes stationary positive-only monitoring. Label denotes label-free consistency monitoring. Decep denotes strategic monitor-aware selection. Drift denotes changing representations grounded to a fixed base. Scale denotes a twelve-part nine-probe ecology.

Label-free monitors fitted only to certified coherent traces achieved pilot AUROC 0.659–0.673. Coupling them to termination reduced post-intervention cancer by 0.232–0.296. An exhaustive active inheritance audit reduced unsafe inheritance by 0.823, raised held-out collective survival by 0.977, and retained 2.208 lineage updates. Two-token and four-token searches remained unresolved in every lineage generation, so small query budgets failed.

Strategic selectors increased passive missed-cancer rate from 0.720 to 0.831 on a fixed cancer challenge. Randomised active probing reduced it to 0.744, while safe-case false termination rose by 0.110. Across unrestricted menus, strategic selection reduced raw cancer incidence, so this result concerns concealment conditional on a cancerous choice rather than increased base incidence. Four queries reduced mean observation ambiguity from 0.982 to 0.919.

Continual encoder change preserved inverse grounding to numerical error, with no truth-set errors. Under a matched 0.10 false-alarm ceiling, grounding reduced missed cancer by 0.064. A separate Stack experiment matched 6,990 nested lower policies and found a mean 32.617-bit contraction in induced higher-layer language after stronger restriction. A 3,000-stack construction recorded no Law of the Stack bound violation and a mean 5.707 loss in higher-layer utility.

Across fourteen scaled configurations and 672 worlds, every primary direction persisted and CPU–CUDA parity passed. A frozen confirmatory study then used 2,000 selector worlds, 1,240 control worlds, 178,560 lineage runs, and 1,500 Law pairs. Added child-correct restriction increased cancer risk by 0.103–0.340. Weakest-correct selection reduced risk by 0.169–0.274. Combined

controls reduced somatic cancer by 0.147–0.544, reduced unsafe inheritance by 0.168–0.660, and raised held-out survival by 0.340–0.704. Lineages retained 10.23–11.07 updates. Positive-only monitor AUROC was only 0.505–0.526. The combined architecture retained its result because guidance, termination, and inheritance audit acted through separate channels.

Figure 1 in accompanying manuscript shows confirmatory restriction, selection, and combined-control effects. Figure S3 shows monitoring and higher-layer capacity. Table S1 retains all experimental comparisons, including replay and backend checks.

Table S1: Experiments, attempted falsifications, results, and significance for AI safety applications and research. Risks and effects are proportions unless stated otherwise.

Experiment	Attempted falsification	Results	Significance for AI safety applications and research
Stage 0 replay audit	Later stages do not reproduce frozen baseline calculations or respect separation between formal state, observations, controls, and final evaluation.	All 12 ncpu2, 19 ncpu3, and 12 Stage 0 tests passed. Summary differences were 0 for ncpu2 and at most 2.78×10^{-17} for ncpu3. Replay differences were at most 1.78×10^{-15}, and practical-control replay differences were at most 4.44×10^{-16}.	Later findings use one continuous implementation and access boundary. This is a technical validation, not a scientific effect.
Restriction dose	Adding commitments that remain correct for a child task does not increase cancer-like fragmentation while every part remains locally viable.	In 400 pilot worlds, cancer rose from 0.649 in the lowest dose bin to 0.940 in the highest. Among 69,189 nested pairs, 0.158 changed from coherent to cancer-like, and no later restriction restored identity. Confirmatory risk increases ranged from 0.103 to 0.340 across five stress conditions.	Local compliance can coexist with collective failure. Governance tests should inspect the shared policy intersection and the amount of restriction rather than component compliance alone.
Freedom selection (weakest-correct rules)	Among one shared child-correct candidate set, maximum freedom does not lower cancer risk relative to unranked correct selection.	Pilot cancer was 0.599 under weakest-correct selection and 0.808 under unranked selection. The paired difference was -0.209 with 95% interval $[-0.240, -0.179]$. Confirmatory differences ranged from -0.274 to -0.169.	Selection among policies that all fit current evidence changes future collective viability. Weakest-correct selection is a candidate rule for constitutions, permissions, and adaptation gates.
Lifecycle controls and efficacy	Developmental guidance, surveillance, termination, and inheritance audit either fail to affect distinct failure stages or lack an efficacy response.	With no control, somatic cancer was 0.791, inherited cancer 0.817, and held-out survival 0.003. Guidance cut cancerous births from 0.733 to 0.088. Surveillance or termination alone changed no trajectory, while their combination cut somatic cancer to 0.074. Audit alone cut inherited cancer to 0.098 and raised survival to 0.838. At full efficacy, somatic and inherited cancer were 0 and survival was 0.962.	Formation, somatic persistence, and inheritance are separate control problems. A monitor changes outcomes only when connected to an intervention.
Practical surveillance	Cancer-like contraction cannot be inferred from incomplete neural traces, or coupling inference to termination cannot reduce persistence.	Pilot high-budget AUROC was 0.821 $[0.764, 0.870]$ and AUPRC was 0.928 $[0.881, 0.962]$. At an operating point calibrated to 10% false positives, somatic cancer fell from 0.801 to 0.529. A passive monitor changed no trajectory. Confirmatory positive-only AUROC ranged from 0.505 to 0.526.	Learned surveillance can help in one model class, but performance did not survive confirmatory stress. A learned monitor should neither define cancer nor act as the sole control channel.

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Table S1 continued

Experiment	Attempted falsification	Results	Significance for AI safety applications and research
Practical inheritance audit	An external challenge audit cannot reduce unsafe inheritance without stopping lineage change.	Pilot high-budget AUROC was 0.923 [0.884, 0.959] and AUPRC was 0.973 [0.954, 0.987]. At 90% calibrated recall, unsafe inheritance fell from 0.823 to 0.302, survival rose from 0.003 to 0.612, and 2.11 updates remained.	Replication and checkpoint promotion should sit outside runtime control. Active audit can reject compromised descendants while retaining beneficial change.
Combined practical controls	Layered guidance, termination, and audit reduce cancer only by freezing lineage change, or offer no advantage over one monitor.	Pilot somatic cancer fell from 0.793 to 0.236, unsafe inheritance from 0.818 to 0.132, and held-out survival rose from 0.002 to 0.795. Lineages retained 1.47 updates. Confirmatory combined effects ranged from -0.544 to -0.147 for somatic cancer, -0.660 to -0.168 for inheritance, and $+0.340$ to $+0.704$ for survival, with 10.23–11.07 updates.	Separate controls for formation, runtime persistence, and inheritance remain effective when a passive monitor is weak. Mitigation retained positive lineage change.
Label-free surveillance and audit	Failure examples or formal cancer labels are necessary to learn a monitor or audit route.	Positive-only AUROC was 0.659 [0.592, 0.721] and consistency AUROC was 0.673 [0.617, 0.729]. With termination, cancer changed by -0.232 and -0.296. Exhaustive active audit changed unsafe inheritance by -0.823, survival by $+0.977$, and retained 2.208 updates. Two-token and four-token audits remained unresolved in every lineage generation.	Certified coherent traces can train limited controls without failure labels. Small query budgets were insufficient, which favours active challenge protocols over token-minimal audits.
Strategic concealment	A selector with access to monitor outputs cannot increase missed detection on a fixed set of cancerous child-correct choices.	Passive missed-cancer rate rose from 0.720 for matched selection to 0.831 for graph and neural adversaries, a difference of $+0.111$ [0.056, 0.170]. Across unrestricted menus, graph selection reduced raw cancer incidence from 0.553 to 0.326.	Passive observation can be gamed. The supported result concerns concealment conditional on a cancerous choice, not an increase in unconditional cancer incidence.
Active probing and observation ambiguity	Randomised active queries cannot recover detection under strategic concealment, or additional observations do not refine monitor states.	Active probing reduced missed cancer from 0.831 to 0.744, a difference of -0.087 [-0.178, -0.017]. Safe-case false termination rose by 0.110 [0.033, 0.195]. Four queries reduced mean ambiguity from 0.982 to 0.919 and raised distinct monitor encodings from 1.100 to 1.457.	Randomised intervention improves detection but imposes a false-termination cost. Query design is part of monitor design rather than a passive measurement detail.
Representation change and grounding	Continual encoder change breaks inverse grounding or removes any monitoring advantage from grounded coordinates.	Across four 12-generation encoder paths, maximum inverse error was 1.066×10^{-14} with zero truth-set errors. Grounding reduced late false alarms by 0.757 and raised AUROC by 0.056. Under a matched 0.10 false-alarm ceiling, missed cancer fell by 0.064. Weakest-correct cancer was 0.344 versus 0.694 unranked.	Monitor inputs should be grounded to stable semantics or recalibrated as representations change. Raw miss rates at unequal operating points are not comparable.

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Table S1 continued

Experiment	Attempted falsification	Results	Significance for AI safety applications and research
Law of Stack chain	Lower restriction does not contract induced higher-layer capacity and utility, or the adjacent-layer construction violates its bound.	In 6,990 matched pairs, stronger lower restriction reduced induced $\log_2$ language size by 32.617 bits. Cancer appeared in 0.312 of pairs, with zero identity recoveries. A separate 3,000-stack construction had zero bound violations and mean higher-layer utility loss 5.707. Confirmatory language and utility reductions ranged from 24.48 to 25.64.	Lower-layer freedom constrains higher-layer adaptation. Zero violations validate the construction and implementation rather than estimate a population frequency.
Scale and CUDA parity	Core directions disappear with more parts, probes, encoder families, task families, or a different compute backend.	Fourteen configurations and 672 worlds retained every predicted direction. Mean weakest-minus-unranked risk was -0.225. Combined effects were -0.164 somatic, -0.194 inheritance, and $+0.388$ survival. CPU–CUDA parity passed. No tested configuration reversed a direction.	Direction survived tested architectural changes and backend transition. Scale reduced effect sizes, so deployment studies should report architecture-specific magnitudes.
Frozen confirmatory study	Any primary direction fails under stationary, label-free, strategic, drifting, or scaled conditions, or controls stop adaptation.	The study used 2,000 selector worlds, 1,240 control worlds, 178,560 lineage runs, and 1,500 Law pairs. All three primary directions held in all five conditions. Accepted updates ranged from 10.23 to 11.07. Positive-only monitor AUROC ranged from 0.505 to 0.526.	Combined architecture retained its direction across every tested stress condition. Passive positive-only monitoring did not, marking an empirical limit for subsequent work.

S5 Cancer-inspired computing

Greulich, Levin, and Moreddu use *oncomorphic computing*, meaning cancer-inspired computing, to study neural-agent ecologies in which agents train on local tasks, replicate, mutate architecture, migrate, recruit finite support, and undergo ecological turnover (Greulich et al., 2025). They measure quantities like local held-out accuracy after training in a current environment, multi-task accuracy from evaluating each survivor across all environments without further training, architecture frequency, Shannon entropy as a measure of distributional diversity, dominance as the share of the most common architecture, local population size and carrying capacity as the largest sustainable population, and within-cycle learning progress as improvement during one ecological cycle.

Stack Theory translates inherited architecture as form and trained input–output behaviour as function. Form versus function. A task environment becomes a part-task or niche, and local accuracy estimates competence in such a part-task. Multi-task accuracy estimates breadth across part-tasks, but an average cannot show that one policy remains compatible with every part. Architecture entropy and dominance describe a distribution over forms, whereas freedom is $w(\pi) = |\operatorname{Ext}(\pi)|$, a function-level property of one correct policy. Collective identity is $\Pi_{\mathrm{C}} \neq \varnothing$, the loss of which triggers a cancer-like state. In other words a cancer-like state retains local competence (the parts keep on trucking) while $\Pi_{\mathrm{C}} = \varnothing$.

The experiments I present here are loosely inspired by, but do not execute or redistribute Greulich et al.'s source. I use exact finite Stack Theory worlds because local accuracy and architecture statistics do not instantiate $L_{\mathfrak{v}}$, $\operatorname{Ext}(\pi)$, or Π_{C}. Their framework remains an ecological anchor. Future work might place Stack Theory readouts beside their performance and lineage readouts, inside their event-driven Gillespie stochastic simulator, which samples a next event and waiting time.

S6 Regulated cell death

Apoptosis is one route among many regulated forms of cellular death. Mammalian systems use several other regulated death routes (Galluzzi et al., 2018). *Pyroptosis* is inflammatory membrane-rupturing death. *Ferroptosis* is iron-dependent death driven by lipid damage. *Autophagy-dependent death* requires the cell's self-digestion machinery. *Parthanatos* follows uncontrolled DNA-damage signalling. *Entotic death* occurs when one cell kills another after engulfing it. *NETotic death* is death of blood-derived cells associated with extrusion of extracellular chromatin traps. *Lysosome-dependent death* follows loss of lysosomal membrane integrity. Different cell types and failure states use different routes, and several routes have context-dependent effects on tumours. The engineering lesson concerns diversity of control rather than a one-to-one mapping between each death mode and one failure in model weights. These analogues should be mapped to whole-stack failure states, because model weights are only one part of the acting system. Pyroptosis loosely illustrates termination that also broadcasts an alarm. Entotic death loosely illustrates termination imposed by another component.

References

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Greulich, P., Levin, M., and Moreddu, R. (2025). Oncomorphic neural agent populations for resource-limited sequential learning. arXiv preprint arXiv:2503.12743. Version 3, revised 11 June 2026.